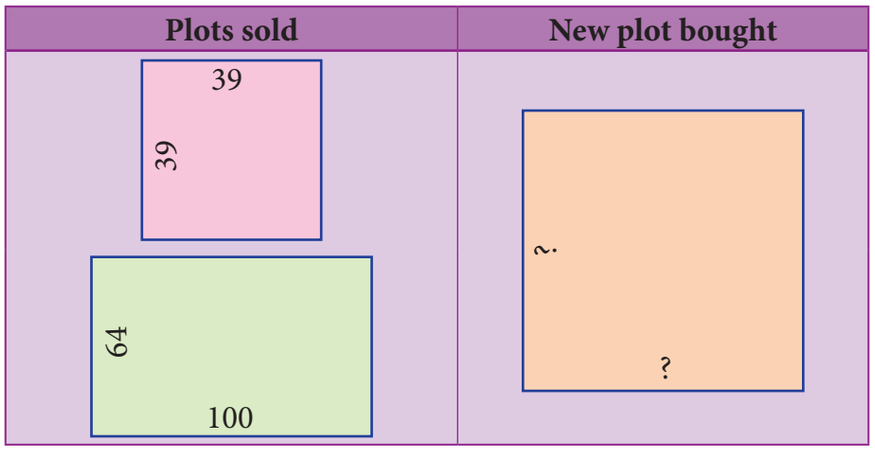
Solution:

The transactions can be visualised as follows:



Area of the square plot bought	=	Area of the square plot sold	+	Area of the rectangular plot sold
	=	39×39	+	100×64
	=	1521	+	6400
	=	7921 m^2		

Length of a side of the new square plot = $\sqrt{7921}$ = 89 m

```
      8  9
    --------
 8  | 79  21
      64
    --------
169 | 15  21
      15  21
    --------
        0
```

1.7 Square Root

Squaring a number is another mathematical operation just like addition, subtraction, multiplication etc., Most mathematical operations have 'inverse' (meaning 'opposite') operations. For example, subtraction is the inverse of addition, division is the inverse of multiplication etc., Squaring also has an inverse operation namely finding the Square root.

The square root of a number n, written $\sqrt{n}$ (or) $n^{\frac{1}{2}}$, is the number that gives n when multiplied by itself. For example, $\sqrt{81}$ is 9, because $9 \times 9 = 81$.

In the adjacent table, we have square roots of all the perfect squares starting from 1 to 100.

If $11^2 = 121$, what is $\sqrt{121}$? If $529 = 23^2$, what is the square root of 529? If we know that $324 = 18^2$, we can immediately tell that $\sqrt{324}$ is 18.

We have, $1^2 = 1$ and so 1 is a square root of 1. Similarly, $(-1)^2 = 1$. So, (-1) is also a square root of 1.

$2^2 = 4$ and so 2 is a square root of 4. Similarly, $(-2)^2 = 4$. So, (-2) is also a square root of 4.

This continues as $3^2 = 9$ and so 3 is a square root of 9. Similarly, $(-3)^2 = 9$. So, (-3) is also a square root of 9.

Square root	Reason
$\sqrt{1} = 1$	$1^2 = 1$
$\sqrt{4} = 2$	$2^2 = 4$
$\sqrt{9} = 3$	$3^2 = 9$
$\sqrt{16} = 4$	$4^2 = 16$
$\sqrt{25} = 5$	$5^2 = 25$
$\sqrt{36} = 6$	$6^2 = 36$
$\sqrt{49} = 7$	$7^2 = 49$
$\sqrt{64} = 8$	$8^2 = 64$
$\sqrt{81} = 9$	$9^2 = 81$
$\sqrt{100} = 10$	$10^2 = 100$

01_8th_Maths_Numbers_Chap 1.indd 28 2/12/2020 12:04:48 PM